GUIDELINE

Delirium Assessment and Management

Scope (Staff):	Medical, nursing and allied health staff
Scope (Area):	Paediatric Critical Care (PCC) unit only

Child Safe Organisation Statement of Commitment

CAHS commits to being a child safe organisation by applying the National Principles for Child Safe Organisations. This is a commitment to a strong culture supported by robust policies and procedures to reduce the likelihood of harm to children and young people.

This document should be read in conjunction with this disclaimer

Contents

Aim

To provide guidance in the prevention, assessment and management of paediatric delirium in the PCC

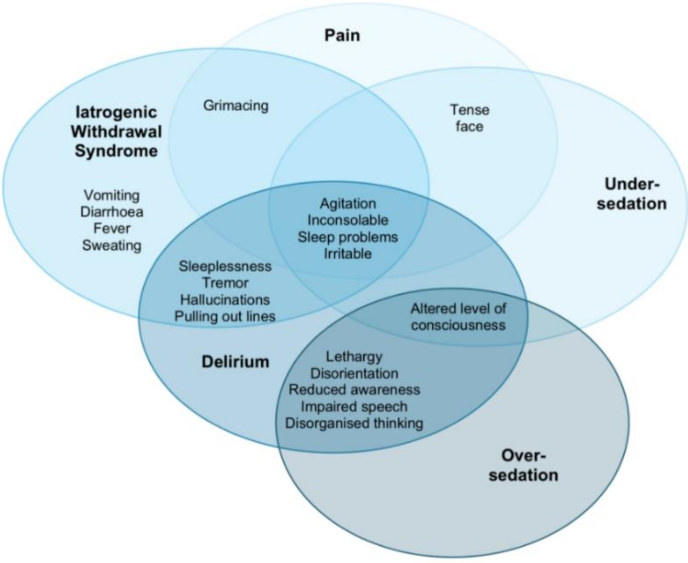
Risk

- Failure to recognise, prevent and treat patients with delirium may result in multiple adverse outcomes including[1-4]
  - increased intensive care unit (ICU) and hospital length of stay (LOS)
  - increased duration of mechanical ventilation
  - increased mortality
  - increased morbidity including pressure injuries, falls, regression of developmental milestones, accidental line removal and long-lasting neurocognitive effects.

Background

- National Safety and Quality Health Service Standard (NSQHS) Standard 5 Comprehensive Care (second edition) highlights cognitive impairment including delirium as a specific area for harm or potential harm in healthcare organisations.[5]
- Action 5.29 of the NSQHS Standards states health service organisations have a system in place for patients at risk of developing delirium which incorporates best-practice strategies for early recognition, prevention, treatment and management of cognitive impairment including delirium.[5]
- Delirium is a common complication of childhood illness, with a reported incidence of up to 29% in children admitted to ICU but it is commonly underdiagnosed.[1,6,7]
- Many of the symptoms of paediatric delirium overlap with pain, withdrawal, and under-sedation and over-sedation (see figure 1). Reliable diagnosis of delirium needs to consider children's developmental stages using a reliable and valid tool.[6]

Figure 1: Overlap of paediatric delirium with pain, iatrogenic withdrawal syndrome, over-sedation and under-sedation.[6]



Definitions

Delirium: State of acute neurologic dysfunction in the context of a critical illness which occurs as a result of the patient's medical condition and/or treatment.[6]

There are 3 main subtypes: hyperactive, hypoactive and mixed-type delirium.

- Main features of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) core diagnostic criteria.[9]
  a) Acute alteration or fluctuation from baseline mental status
  b) Inattention
  c) Altered level of consciousness
  d) Disorganised systems / thinking

Hypoactive delirium: Is characterised by apathy, reduced awareness, lethargy, inattentiveness, slowed or sparse speech, or slowed motor activity.[6,8] This is the most common type and has the poorest prognosis.[8]

Hyperactive delirium: Is characterised by irritability, purposeless actions, agitation, restlessness, hypervigilance and combative behaviours that do not improve with adequate pain relief.[6,8]

Mixed delirium: Child fluctuates between both a hypoactive and hyperactive state. [6,8] This is the second most common type.[6,8]

Key points

- The prevention, screening and management of patients with delirium is summarised in the PCC Delirium Clinical Pathway, see Appendix 1.

- Delirium prevention should occur for all patients admitted to PCC – in particular through maintenance of day-night routine, sleep promotion, familiarisation of environment.
- Day and night routine promotes high quality sleep and the maintenance of a normal circadian rhythm. Sleep deprivation or poor-quality sleep is a significant risk factor in the development of delirium.
- The Cornell Assessment of Paediatric Delirium (CAPD) is a reliable, validated and rapid observational tool to detect delirium in critically unwell children of all ages and developmental stages and is recommended to assess paediatric delirium.[6,10,11] (see Appendix 2)

Nursing care

- Delirium prevention measures should be instituted as a standard for all PCC patients to help minimise the incidence of delirium.[12]
- Preventative measures include:[11-13,]
    - Avoidance of deep sedation.
        - In intubated children, titrate sedation to target State Behavioural Scale (SBS).
        - In non-intubated children, notify medical staff for medication review if they appear over-sedated.
    - Provide adequate analgesia
        - Target Multidimensional Assessment of Pain Scale (MAPS) of <4 in all intubated patients
        - In non-intubated children use an age appropriate pain score such as the revised Faces, Legs, Activity, Cry, Consolability (rFLACC) score (up to 8 years) or self-reporting via the Faces Pain Scale (> 4 years) or the Numeric pain rating scale (> 8 years).
        - see analgesia and sedation guideline for more information
    - Aim for spontaneous breathing on ventilator where appropriate
    - Re-orientate patient to time and place regularly (as appropriate for age)
    - Promote a day-night routine (see Appendix 4)
    - Promote family involvement
    - Promote a familiar environment as much as possible (encourage parents to bring in toys, photos, comfort objects)
    - Ensure access to communication aids (e.g. glasses / hearing aids / tablets)
    - Daily review of device removal (e.g. lines, tubes)
    - Encourage early mobilisation as appropriate

- Provide parents / caregivers of patients at risk of delirium with a 'Delirium in children and adolescents' Health Fact sheet (see risk factors in medical care section below).
- Perform CAPD scoring every 12 hours from the time of admission for all patients admitted to PCC (see Appendix 2).
    - If child is <2 years of age, then the developmental anchors should be used to assist in the CAPD scoring.[11,13] (see Appendix 3)
    - Where possible, involve patient's parents / caregivers in screening as they know their child best. This is particularly important when assessing children with pre-existing neurocognitive delays or developmental disabilities.
    - A score ≥ 9 is a positive screen for delirium. Alert medical staff to review patients for assessment using BRAIN MAPS as per clinical pathway in Appendix 1.
    - A score of < 9 indicates that the patient does not have delirium. Continue to reassess every 12 hours until discharge from PCC.

Medical care

- Have a high index of suspicion of delirium in all children admitted to PCC, particularly for hypoactive delirium which can be harder to detect.
- Risk factors for delirium in children include:[1,6,8]
    - Severe illness – particularly severe infective and inflammatory conditions
    - Age less than 5 years (especially less than 2 years)
    - Pre-existing cognitive impairment or developmental delay
    - Mechanical ventilation
    - Pre-existing medical condition, in particular metabolic and neurological disorders
    - Cardiopulmonary bypass
    - Exposure to medications, in particular benzodiazepines, ketamine and polypharmacy
    - Length of stay in PCC >24 hours
- All patients with a CAPD score ≥ 9 require a medical review to assess for differential diagnoses and potential causes and contributors to delirium.
- Delirium is a diagnosis of exclusion and differential diagnoses should always be considered including:
    - Over-sedation with altered level of consciousness[6,8]
    - Iatrogenic withdrawal syndrome (see Withdrawal Syndrome Management guideline).[6,8,13]

- Anticholinergic toxidrome, in particular overlapping features of psychiatric behaviour, motor state and confusion.[13]
- Evolving infection.[8,12]
- Hypoxia.[8,12]
- New organ dysfunction (including central nervous system (CNS), cardiovascular (CV), hepatic, renal and endocrine dysfunction).[8,12]
- Acute metabolic derangements.[8,12]

- The BRAIN MAPS tool below should be used to work through the common causes and contributors to delirium.[12,14]

Table 1: BRAIN MAPS Tool

	Assessment	Evaluation	Recommendations
B	**Bring oxygen**	Evaluate patient for: • Hypoxaemia • Low cardiac output • Anaemia	Improve oxygen delivery by: • Supplemental oxygen • Packed red blood cell (PRBC) transfusion if anaemic
R	**Remove / reduce drugs**	Evaluate for use of: • Sedatives, especially benzodiazepines • Anticholinergic drugs	Reduce or discontinue if possible
A	**Atmosphere**	Room setup —bright lights, noise levels Restraint use Caregiver presence Schedule / routine Use of adaptive equipment and / or communication aids (e.g. glasses / hearing aids)	Encourage normal day / night routine Encourage family presence Provide for continuity of care in nursing staff as able
I	**Infection / mobilisation / inflammation**	Evaluate patient for: • Infection • Immobilisation • inflammation	Treat fever and infection Encourage mobilisation as able
N	**New organ dysfunction**	Consider all systems: CNS, CV, pulmonary, hepatic, renal, endocrine	Treat organ dysfunction as appropriate
M	**Metabolic disturbance**	Evaluate with Arterial Blood Gas (ABG) / Venous Blood Gas (VBG) for: • Hypo / hypernatremia • Hypo / hyperkalaemia • Hypocalcaemia • Alkalosis/acidosis	Normalise electrolytes
A	**Awake**	No day / night routine Sleep wake cycle disturbance	Establish day / night routine Consider use of melatonin if ongoing disturbances to day-night cycle

	Assessment	Evaluation	Recommendations
P	**Pain**	Untreated or undertreated pain Over-treated (sedated) Check appropriate pain score	Adjust analgesia regime as appropriate; Consider Acute Pain Service (APS) referral
S	**Sedation**	Review benzodiazepine use Assess need for sedation and current sedation target	Weaning / stop benzodiazepines if able Minimise polypharmacy

Pharmacological therapy

- The primary management of delirium is non-pharmacological measures and management of underlying causes (e.g. sepsis, hypoxia, pain, withdrawal).
- Melatonin can be considered for promotion of sleep wake cycle which may improve time to resolution of delirium.
- Use of other pharmacological therapy will not alter the time to resolution of symptoms but can be helpful to manage symptoms to reduce risk of injury to self or others and risk of device dislodgement.
- Baseline electrocardiogram (ECG) should be performed prior to use of atypical antipsychotics.
- In general, clonidine should be the first line medication for agitation. If this is insufficient, then quetiapine is the next line agent.
- If the patient is unable to swallow tablets and does not have a nasogastric tube (NGT) in situ, then olanzapine orally disintegrating tablets (ODT) or wafer can be used as an alternative to quetiapine.
- Droperidol can be used to gain control of an acute behavioural disturbance in children in whom IV clonidine is insufficient and enteral medication is unable to be administered. If droperidol is used, then a nasogastric tube (NGT) should be placed when the acute situation is controlled to facilitate subsequent pharmacotherapy with quetiapine due to a better side effect profile with similar efficacy.[12]
- If pharmacological therapy is commenced, then review daily for suitability of ceasing and continue no longer than 5-7 days.[14]

Table 2: Medications for agitation

Medication	Route	Dose	Side effects and additional information
Clonidine[15] 0-18 years 1st line	Enteral	1 - 4 microg/kg 6 hourly (max 150 microg/dose)	Bradycardia, hypotension, sedation
	IV	1 - 2 microg/kg 6 hourly (max 50 microg/dose)	Bradycardia, hypotension, sedation
Quetiapine[14,16] 6-18 years 2nd line	Enteral	**6-11 yrs:** Initial dose: 12.5 – 25 mg, can repeat hourly until sufficient effect achieved. Subsequent dosing: total initial cumulative dose given daily in 2 divided doses (cumulative max 200 mg/day) **≥ 12 yrs:** Initial dose: 12.5 mg - 50 mg, can repeat hourly until sufficient effect achieved. Subsequent dosing: total initial cumulative dose given daily in 2 divided doses (cumulative max 200 mg/day)	Dystonic reactions, long QT syndrome (LQTS), sedation. Mainly histaminergic effect at lower doses.
Olanzapine[17,18] 6-18 years 3rd line	Enteral / IM	**< 40 kg:** 2.5 - 5 mg. Repeat after 2-4 hours. Max daily dose 10 mg **> 40 kg:** 5-10 mg. Repeat after 2-4 hours. Max daily dose 20 mg	Dystonic reactions, LQTS, sedation. Consider a lower starting dose in renal or hepatic impairment. Enteral – wafer preferred
Droperidol[19] 6-18 years Consider only if unable to administer enteral medication 4th line	IM / IV	0.03 - 0.07 mg/kg (max 5 mg) If necessary, repeat in 30 mins. Maximum 4 doses in 24 hours. ***If droperidol is given then a nasogastric tube should be placed to facilitate subsequent pharmacotherapy with quetiapine.**	Dystonic reactions, tardive and withdrawal dyskinesia, LQTS, hypotension, extrapyramidal side effects, anticholinergic effects. Lower risk of neurotoxicity than haloperidol.

Allied Health Support

Allied Health services should be considered where support is required for assessment and management of communication, orientation to person / place / time, behaviours and/or psychological distress of patient and or primary caregivers.

PCC Physiotherapist

- All patients in PCC have a mandatory referral to the PCC Physiotherapist for assessment of Early Mobility.
- PCC Physiotherapist conducts assessment of mobility and discusses this with medical and nursing staff as well as the patient's family.

Engage referral to PCC Occupational Therapist:

- For supportive equipment when encouraging mobilisation in state of delirium.
- To support opportunities for meaningful stimulation and provide education to family and carers, including benefits of familiar objects and orientation supports.

Engage referral to PCC Speech Pathologist:

- For communication support; visual cues (encouraging alternative communication AAC, pictures and gestures) and cueing systems to support comprehension.
- To monitor risk of aspiration if alertness fluctuates and adjust diet if needed.

Follow up care – Ward Transfer

- All patients must have a CAPD screen completed prior to transfer to another ward. Document the CAPD score on MR852.31 Paediatric Critical Care to Ward Transfer Form.
- Patients with a positive score must be medically reviewed prior to transfer by both PCC and ward medical team to determine appropriateness of ward transfer
- PCC medical staff will inform the accepting treating team of the positive delirium score and management plan. The STARS CNS in their usual duties reviews all PCC discharges within the last 24 hours. The STARS CNS will follow up delirium screening for 24 hours on the ward.
- The STARS CNS will handover patients with a positive delirium score to the Safety Team After-Hours Response Service (STARS) medical staff.
- Ensure all medications that are no longer required for the management of delirium are ceased, or appropriate weaning plans are documented. Provide Delirium Health Facts to patient and caregivers.

Related CAHS internal policies, procedures and guidelines
Withdrawal Syndrome Management guideline (Clinical Practice Manual)
Analgesia and Sedation in PCC guideline (PCC Practice Manual)
Clonidine monograph (PCH website)
Recognising and Responding to Acute Deterioration CAHS
Droperidol (PCH website)

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Useful resources (including related forms)
‘Delirium in children and adolescents’ Health Facts sheet https://pch-healthpoint.hdwa.health.wa.gov.au/directory/CIS/HF/Documents/Delirium%20in%20children%20and%20adolescents.pdf
Critical Illness, Brain Dysfunction and Survivorship centre. https://www.icudelirium.org/
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File Path:	W:\Safety & Quality\CAHS\CLOVERS SSCCU PICU\Procedures Protocols and Guidelines\Word\PCH.PCC.DeliriumAssessmentAndManagement.docx		
Document Owner:	Co Director Surgical Services		
Reviewer / Team:	Kate Douglas (PCC Consultant), Blaise Donnelly (PCC CN), Basia Hamden (A/HOD Speech Pathology), Jess Lupu (Speech Pathologist), PCC S&Q CN, Alexander Thompson (Psychiatry Consultant), PCC Pharmacist, NS5 Comprehensive Care Committee, NS8 Acute Clinical Deterioration Committee		
Date First Issued:	March 2022	Last Reviewed:	Mar 2025
Amendment Dates:	Sept 2025	Next Review Date:	Apr 2028
Approved by:	PCHN Medication Safety Committee	Date:	Jul 2025
Endorsed by:	CAHS Drug & Therapeutics Committee	Date:	Aug 2025
Aboriginal Impact Statemen Declaration		Date:	02.04.2025
Standards Applicable:	NSQHS Standards: Child Safe Standards: 1,10		

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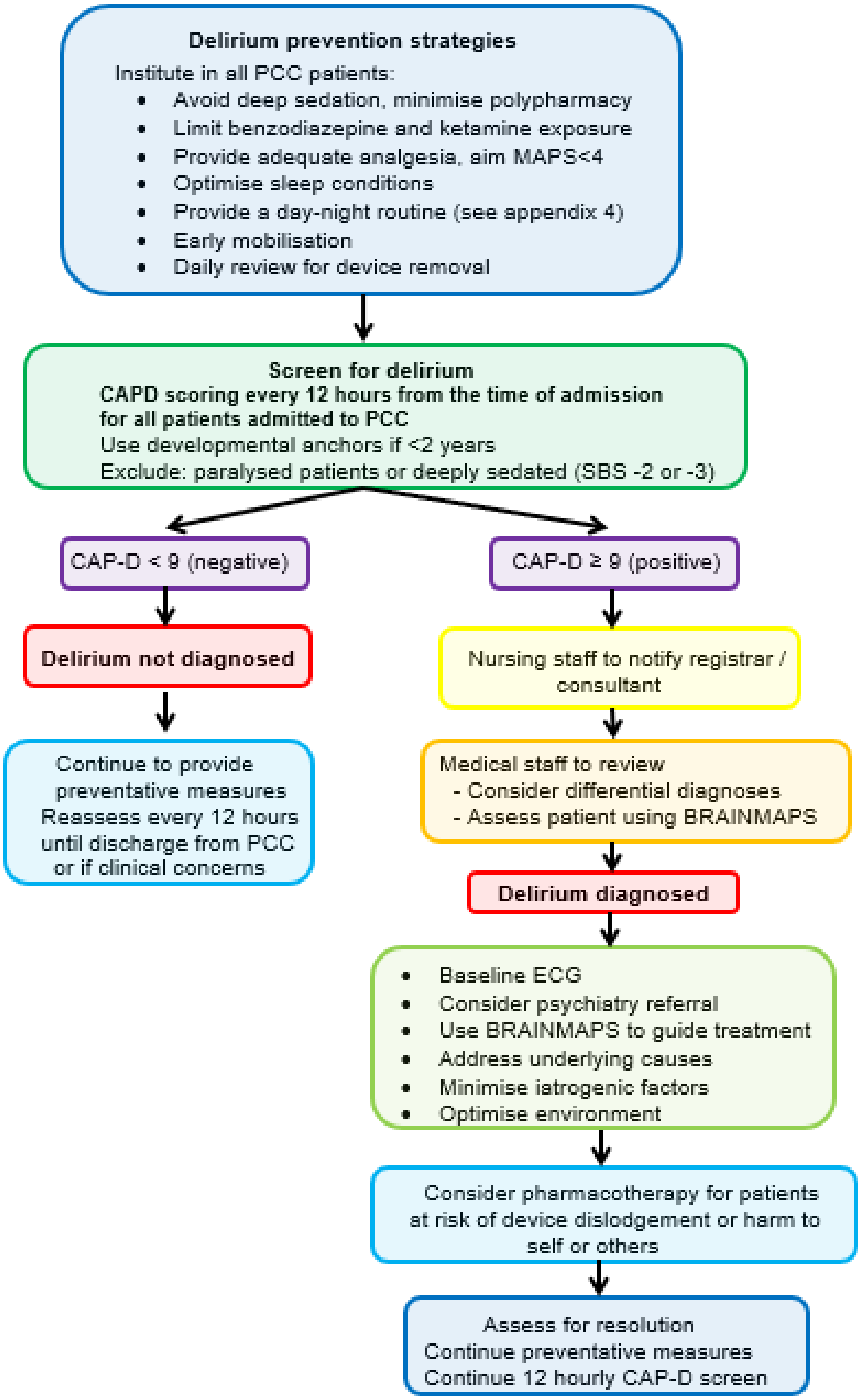
Appendix 1: PCC Delirium Clinical Pathway



Appendix 2: Cornell Assessment of Paediatric Delirium Tool

Please answer the following questions based on your interactions with the patient over the course of your shift						
	Never **4**	**Rarely** **3**	**Sometimes** **2**	**Often** **1**	**Always** **0**	**Score**
Does the child make eye contact with the caregiver?						
Are the child's actions purposeful?						
Is the child aware of his/her surroundings?						
Does the child communicate needs and wants?						
	Never **0**	**Rarely** **1**	**Sometimes** **2**	**Often** **3**	**Always** **4**	**Score**
Is the child restless?						
Is the child inconsolable?						
Is the child underactive: very little movement and interaction?						
Does it take the child a long time to respond to interaction?						
					Total	

Appendix 3: Developmental Anchor Points for Youngest Patients

	NB	4 weeks	6 weeks	8 weeks	28 weeks	1 year	2 years
1. Does the child make eye contact with the caregiver?	Fixates on face	Holds gaze briefly Follows 90 degrees	Holds gaze	Follows moving object/caregiver past midline, regards examiner's hand holding object, focused attention	Holds gaze. Prefers primary parent. Looks at speaker	Holds gaze. Prefers primary parent. Looks at speaker	Holds gaze. Prefers primary parent. Looks at speaker
2. Are the child's actions purposeful?	Moves head to side, dominated by primitive reflexes	Reaches (with some discoordination)	Reaches	Symmetric movements, will passively grasp handed object	Reaches with coordinated smooth movement	Reaches and manipulates objects, tries to change position, if mobile may try to get up	Reaches and manipulates objects, tries to change position, if mobile may try to get up and walk
3. Is the child aware of his/her surroundings?	Calm awake time	Awake alert time Turns to primary caretaker's voice May turn to smell of primary care taker	Increasing awake alert time Turns to primary caretaker's voice May turn to smell of primary care taker	Facial brightening or smile in response to nodding head, frown to bell, coos	Strongly prefers mother, then other familiars. Differentiates between novel and familiar objects	Prefers primary parent, then other familiars, upset when separated from preferred care takers. Comforted by familiar objects especially favorite blanket or stuffed animal	Prefers primary parent, then other familiars, upset when separated from preferred care takers. Comforted by familiar objects especially favorite blanket or stuffed animal
4. Does the child communicate needs and wants?	Cries when hungry or uncomfortable	Cries when hungry or uncomfortable	Cries when hungry or uncomfortable	Cries when hungry or uncomfortable	Vocalizes /indicates about needs, eg. hunger, discomfort, curiosity in objects, or surroundings	Uses single words, or signs	3-4 word sentences, or signs. May indicate toilet needs, calls self or me
5. Is the child restless?	No sustained awake alert state	No sustained calm state	No sustained calm state	No sustained calm state	No sustained calm state	No sustained calm state	No sustained calm state
6. Is the child inconsolable?	Not soothed by parental rocking, singing, feeding, comforting actions	Not soothed by parental rocking, singing, feeding, comforting actions	Not soothed by parental rocking, singing, feeding, comforting actions	Not soothed by parental rocking, singing, comforting actions	Not soothed by usual methods eg. singing, holding, talking	Not soothed by usual methods eg. singing, holding, talking, reading	Not soothed by usual methods eg. singing, holding, talking, reading (May tantrum, but can organize)
7. Is the child underactive—very little movement while awake?	Little if any flexed and then relaxed state with primitive reflexes (Child should be sleeping comfortably most of the time)	Little if any reaching, kicking, grasping (still may be somewhat discoordinated)	Little if any reaching, kicking, grasping (may begin to be more coordinated)	Little if any purposive grasping, control of head and arm movements, such as pushing things that are noxious away	Little if any reaching, grasping, moving around in bed, pushing things away	Little if any play, efforts to sit up, pull up, and if mobile crawl or walk around	Little if any more elaborate play, efforts to sit up and move around, and if able to stand, walk, or jump
8. Does it take the child a long time to respond to interactions?	Not making sounds or reflexes active as expected (grasp, suck, moro)	Not making sounds or reflexes active as expected (grasp, suck, moro)	Not kicking or crying with noxious stimuli	Not cooing, smiling, or focusing gaze in response to interactions	Not babbling or smiling/laughing in social interactions (or even actively rejecting an interaction)	Not following simple directions. If verbal, not engaging in simple dialogue with words or jargon	Not following 1-2 step simple commands. If verbal, not engaging in more complex dialogue

Appendix 4: Day and Night Routine in PCC

Day time	Rationale
Lighting • Main PCC lights on 0700 • Patient bedspace lights on 0700 • Raise all window blinds	• Promote normal circadian rhythm by providing day light • May help patient sleep better at night
Support usual day time nap schedule	• Maintain patient's normal sleeping patterns
Provide meaningful stimulation Provide education to family and carers re: rationale for familiar objects and orientation supports • Patients own toys • Comfort objects (blankets, dummy) • Photographs of family • Books • Mobiles / pictures • Clocks visible • Orientate patient	• Creates familiar environment • May be reassuring to patient • Orientates patients to time, person, place, reduces risk of ICU delirium
Provide vision and hearing aids if used	• Orientates patients to time, person, place and reduces risk of delirium
Encourage mobilisation • Sitting up in bed • Getting patient out into chair • Cuddles with family Physiotherapy involvement OT involvement re: supportive equipment	• Promotes a day night routine • Decreases ICU-acquired weakness, reduces risk of ICU delirium
Involve caregivers and family	• Orientates patient, may reduce ICU delirium

Night time	Rationale
Designated sleeping time 2000 - 0700	• Coincides with nursing shift hours • Promotes normal circadian rhythm
Minimise night-time noise • Turn off TV (unless used for lullabies) • Lower staff conversations • Ask visitors to lower conversations • Reduce visitors after 1900 other than parent / guardians • No non-clinical conversations around patient bedspace • Respond to alarms as soon as possible	• Decreases interruptions to sleep. • Promotes normal circadian rhythm • Promotes uninterrupted sleep, allowing child to cycle through normal sleep cycles
Minimise night-time lighting • Dimming corridor, staff station lights and patient lights • Night mode on ventilator • Face pumps away from patients • Offer eye covering • Close blinds and curtains • Television off	• Decreases interruptions to sleep. • Promotes normal circadian rhythm • Promotes uninterrupted sleep, allowing child to cycle through normal sleep cycles
Cluster nursing cares Cluster nursing cares 4 hourly where possible or coincide with medication times or when child wakes	• Cluster cares – doing all hands-on care at the same time to decrease frequency of disruptions to sleep. • Allow for periods for uninterrupted sleep
Patient comfort • Adequate pain relief • Positioning – preferred sleeping position, pillows, position of lines • Body temperature	• Adequate pain relief supports sleep. • Reduces risk of ICU delirium • Ensure patient comfort to promote sleep. • Record preferred sleeping position on PCC nursing care plan